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Enhanced Soil Fertility Classification Using ACO-Optimized XgBoost and Explainable AI

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Abstract. This work presents a soil classification model that includes both micro and macro soil nutrients for soil fertility classification. A hybrid machine learning model is proposed, where the hyperparameters of the eXtreme Gradient Boosting (XgBoost) algorithm are optimized using the Ant Colony Optimization (ACO) algorithm to enhance the model's performance. The ACO-XgBoost hybrid model is benchmarked against traditional machine learning algorithms, achieving an impressive accuracy of 98%. The best-performing ACO-XgBoost model is further analyzed using Explainable AI (XAI) techniques to provide interpretability and understanding of the model's decision-making process. This analysis shows that important soil properties help in identifying soil fertility correctly.

Keywords: Soil classification, eXtreme Gradient Boosting (XgBoost), Ant Colony Optimization (ACO), Hybrid machine learning, Explainable AI (XAI)

1 Introduction

In India, farming is the primary basis of living; about 70% of rustic families depend on farming for their income [1, 2]. India is the major producer and consumer of agricultural products, accounting for 25% of universal production and 27% of universal consumption [3]. Understanding the chemical, physical, and biological characteristics of soil is vital for the productivity of agriculture. It is crucial to increase soil fertility according to its chemical properties in order to increase crop yield. However, soil deterioration, like erosion, compaction, nutrient depletion, and contamination, causes a reduction in soil fertility. Crops grown in nutrient-deficient or nutrient-excess soils can experience diseases that lead to poor productivity. Therefore, accurately assessing soil parameters is vital for improving yields and ensuring sustainable farming practices that do not harm the environment.

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Soil fertility is influenced by several aspects, together with pH, organic content, cation exchange capacity (CEC), nutrient proportions, and soil texture. Soils with low pH have poor nutrient retention, leading to reduced fertility. The key soil chemical parameters necessary for plant growth are categorized into non-mineral and mineral nutrients. Non-mineral nutrients, such as carbon, oxygen, and hydrogen, are obtained from air and water. Mineral nutrients are divided into macronutrients (nitrogen, potassium, phosphorus, calcium, sulfur, and magnesium) and micronutrients (boron, manganese, chloride, iron [4], copper, molybdenum, zinc, and nickel). They play critical roles in various growth stages of crops, such as seed germination, root and leaf growth, photosynthesis, and disease confrontation. Their imbalances can adversely affect the crop growth and yield.

Machine learning methods have become increasingly popular in various domains and are widely used for prediction and classification tasks. Despite their extensive applications, there is limited research on the use of machine learning for predicting soil fertility. This work estimates the performance of machine learning classifiers for soil fertility classification, aiming to enhance the accuracy and efficiency of soil management practices. The main contributions of this work are as follows:

- A soil classification model is proposed, which utilizes both micro and macro soil nutrients for accurate categorization.
- In this work, we present a hybrid machine learning model in which hyperparameters of the XgBoost algorithm are improved with the Ant Colony Optimization (ACO) algorithm [5] to achieve better classification performance.
- The hybrid model ACO-XgBoost gives a high accuracy (98%) when compared with the conventional machine learning models.
- To provide more understanding, the best-performing model (ACO-XgBoost) is investigated using Explainable AI (XAI) techniques to highlight the most important features contributing to soil fertility classification.

2 Literature Survey

Behera et al. [6] designed an IoT-based system for soil health nursing, using wireless sensor networks to collect data from the field. Their system was able to track soil properties continuously and transmit the data to a cloud server for further scrutiny, demonstrating its potential for scalable soil fertility assessments in precision agriculture. This integration of IoT increases the accuracy and timeliness of soil fertility classification, offering farmers actionable insights for optimized fertilization practices. Agarwal et al. [7] proposed a system for determining soil fertility using colorimetry, where a soil sample is dissolved in an aqueous solution and the intensity of the resulting color is measured using a color sensor. The intensity values are classified into three categories such as low, medium, and high using the Naïve Bayes classification algorithm, providing an efficient and accurate method for soil nutrient analysis. The authors demonstrated that their system significantly reduces the time required for soil fertility testing compared

to traditional methods, making it a practical tool for farmers to quickly assess soil health. Natsir et al. [8] proposed a method for classifying agricultural soil fertility levels based on soil texture using Convolutional Neural Networks (CNN). The study utilized a dataset of 1400 soil images, divided into sand, clay, and sand-clay categories, and achieved an accuracy of 94.24% after tuning CNN parameters such as learning rate, optimizer, and batch size. The authors demonstrated that CNNs, with their capacity to retain spatial information from images, significantly outperformed traditional soil fertility classification techniques. Sunori et al. [9] proposed the use of Learning Vector Quantization (LVQ) and Probabilistic Neural Network (PNN) models to classify soil fertility based on pH and potassium concentration values. The authors developed both models using MATLAB and conducted performance comparisons, with PNN achieving a 100% classification accuracy, while the LVQ model reached 91.66%. This work demonstrates the effectiveness of ANN-based models in accurately predicting soil fertility and provides valuable insights into the potential applications of machine learning in soil management. M et al. [10] proposed the use of several machine learning classifiers, including Naïve Bayes, logistic regression, Support Vector Machine (SVM), decision tree bagging, Boosted Regression Tree (BRT), and Random Forest (RF), for classifying soil fertility levels into low, medium, and high categories based on chemical soil parameters. The authors found that the RF classifier outperformed the other models, achieving the highest accuracy (99.9978%) and a kappa statistic of 1. These findings demonstrate the effectiveness of machine learning-based approaches in enhancing soil fertility prediction and decision-making in agriculture. Sujatha and Jaidhar [10] proposed a Convolutional Neural Network (CNN)-based approach for classifying soil fertility levels as HIGH, MEDIUM, or LOW based on chemical measurements of soil parameters such as pH, EC, and nutrients like P, K, and Zn. The authors achieved an accuracy of 97.24% using an 11×11 input grid and 3×3 kernel size, and further improved performance with the Synthetic Minority Oversampling Technique (SMOTE), reaching an accuracy of 97.52%. This model not only classifies soil fertility but also provides fertilizer recommendations tailored to the specific nutrient deficiencies identified through the classification.

The reviewed literature highlights various advancements in soil fertility classification using machine learning and deep learning techniques, but a common limitation across these studies is the lack of exploration into explainable AI (XAI). While many of the papers propose methods that successfully classify soil fertility based on chemical and physical soil parameters, none of them delve into identifying and explaining the key factors influencing the classification process, which is crucial for making the results more interpretable and actionable for farmers. Furthermore, although a variety of algorithms, including Convolutional Neural Networks (CNN), Random Forest (RF), and Naïve Bayes, have been used to enhance classification accuracy, there is no mention of hybrid models combining these techniques with XAI approaches to not only improve accuracy but also offer transparency. A promising direction would be the development of a new hybrid algorithm that incorporates explainable AI, providing both high per-

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formance in soil fertility classification and clear insights into the critical factors, ultimately supporting more efficient and trustworthy decision-making in precision agriculture.

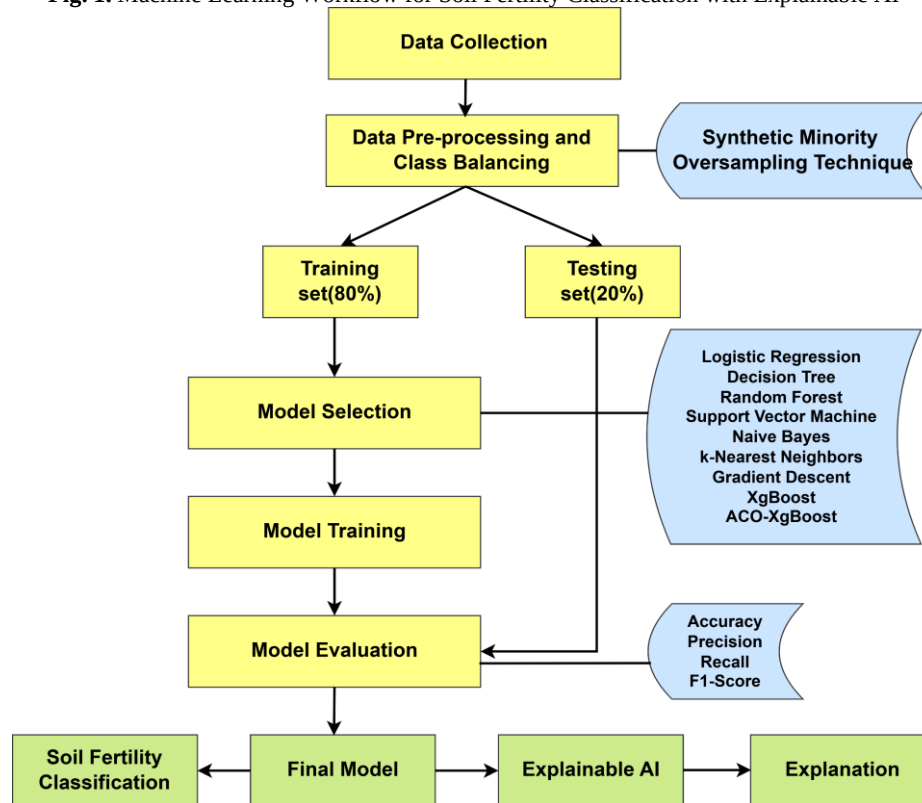
2.1 Reasons for Soil Fertility Classification

Soil fertility classification is vital for precision agriculture as it gives optimized resource management, increased crop yields, and sustainable farming practices. By accurately assessing soil nutrient levels, farmers can apply fertilizers and other resources precisely where needed, by reducing waste and lowering costs. This targeted approach not only boosts productivity but also promotes environmental sustainability by preventing over-fertilization. Additionally, soil fertility data supports data-driven decisions and integrates seamlessly with other precision farming technologies, enhancing overall farm management. Ultimately, it helps maintain long-term soil health and ensures the continued viability of agricultural practices.

3 Methodology

The Fig. 1 represents the workflow for the soil fertility classification system. The process begins with data collection, data collected from the research paper [11], including features such as N: Nitrogen content, P: Phosphorus content, K: Potassium content, pH: Soil pH, EC: Electrical conductivity, OC: Organic carbon, S: Sulfur content, Zn: Zinc content, Fe: Iron content, Cu: Copper content, Mn: Manganese content, B: Boron content. Once the data is collected, it undergoes data preprocessing, which involves steps like cleaning the dataset by removing irrelevant columns, handling missing values, and applying normalization to scale the features to a consistent range. This step balances the dataset classes using SMOTE, making equal class distribution. After balancing, each class contains exactly 440 data points. The training and testing split for each class are as follows: for class 0, the ratio is 356:84; for class 1, it is 347:93; and for class 2, it is 353:87. Model training is then performed, where ML algorithms such as Logistic Regression[4], Decision Trees[4], Random Forest[4], Support Vector Machines[4], Naive Bayes, k-Nearest Neighbors, Gradient Descent, eXtreme Gradient Boosting, and ACO-eXtreme Gradient Boosting are applied to train the model. After training, the model undergoes evaluation using various performance metrics, i.e., accuracy, precision, recall, and F1-score. These metrics are used to evaluate its classification capability. Conclusively, the system is capable of making soil fertility classification and generating forecasts based on new input data. Once the final model is selected, an explanation of how the model makes decisions is provided, which helps in understanding the predictions made by the model and ensures trust and reliability in its use.

Fig. 1. Machine Learning Workflow for Soil Fertility Classification with Explainable AI



3.1 Dataset Description

The dataset used for soil fertility classification consists of various soil properties including Nitrogen (N), Phosphorus (P), Potassium (K), pH, Electrical Conductivity (EC), Organic Carbon (OC), Sulfur (S), Zinc (Zn), Iron (Fe), Copper (Cu), Manganese (Mn), and Boron (B), along with an output class representing soil fertility (class 0, class 1, class 2). [11]

3.2 Machine Learning Algorithm

In this study, various machine learning (ML) algorithms, including Logistic Regression, Decision Trees, Random Forest, Support Vector Machines, Naive Bayes, k-Nearest Neighbors, Gradient Descent, eXtreme Gradient Boosting (XgBoost), and Ant Colony Optimization-eXtreme Gradient Boosting (ACO-XgBoost), are applied to train the model. Each of these models was evaluated based on their performance, and it was found that XgBoost delivered the best results. To further enhance its performance, the

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hyperparameters of the XgBoost model are optimized using ACO. This hybrid approach, combining the strengths of both XgBoost and ACO (shown in Algorithm 1, led to improved classification accuracy and efficiency, achieving better results compared to other models in the study.

Algorithm 1 Hybrid XgBoost with Ant Colony Optimization for Soil Fertility Classification

- 1: **Input:** Soil properties $x_i \in \mathbb{R}^{12}$ (N, P, K, pH, EC, OC, S, Zn, Fe, Cu, Mn, B) to predict soil fertility class $y \in \{0, 1, 2\}$.
- 2: **Output:** Optimized hyperparameters h^* and trained XgBoost model.
- 3: Initialize pheromone levels τ_{ij} , number of ants m , iterations N_{iter} , evaporation rate ρ , and influences α, β .
- 4: **for** each iteration $t = 1 \dots N_{iter}$ **do**
- 5: **for** each ant $A_k, k = 1 \dots m$ **do**
- 6: **Step 1: Construct Solution**
- 7: **for** each hyperparameter $h_i \in h$ **do**
- 8: Select h_j based on $P(h_j|h_i) = \frac{\tau_{ij}^\alpha \cdot \eta_{ij}^\beta}{\sum_{j'} \tau_{ij'}^\alpha \cdot \eta_{ij'}^\beta}$
- 9: **end for**
- 10: **Step 2: Evaluate Solution**
- 11: Train XgBoost with h_k , compute validation error $\mathcal{L}_k$
- 12: Update h_{best} if $\mathcal{L}_k < \mathcal{L}_{best}$
- 13: **end for**
- 14: **Step 3: Update Pheromone**
- 15: **for** each hyperparameter pair (i, j) **do**
- 16: Update $\tau_{ij} \leftarrow (1 - \rho) \cdot \tau_{ij} + \frac{1}{\mathcal{L}_k}$
- 17: **end for**
- 18: **end for**
- 19: **Step 4: Return Best Hyperparameters h^***
- 20: **Step 5: Retrain Final Model with h^***
- 21: **Step 6: Classify Fertility $\hat{y}_i = \text{XgBoost Classifier}(x_i|h^*)$**

4 Results and Discussion

Table 1 highlights the outstanding performance of ACO-XgBoost, consistently outperforming other models across all three classes. Achieving an accuracy of 0.98, ACO-XgBoost surpasses models like XgBoost (0.93), Gradient Descent (0.92), and Random Forest (0.93). ACO-XgBoost also dominates precision between 0.97 and 0.98, followed closely by XgBoost and Gradient Descent (0.93). The recall follows the same trend with ACO-XgBoost, achieving 0.97 to 0.98. At the same time, models like Naive Bayes struggle, particularly in Class 2, with a recall of just 0.48. Additionally, ACO-XgBoost maintains an excellent f1 score of 0.98 across all classes. XgBoost and Gradient Descent perform well but do not match the overall balanced performance of ACO-XgBoost. Furthermore, conventional models such as Logistic Regression, Naive Bayes, and Support Vector Machine show weaker results, particularly in recall and F1-score.

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Overall, ACO-XgBoost is the most effective model for this classification task, demonstrating robust performance across all key metrics.

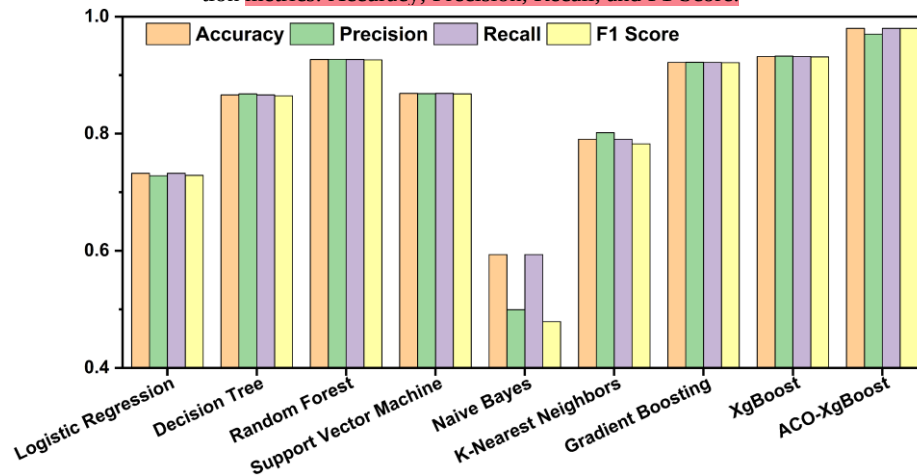
	Accuracy	Precision	Recall	F1-Score	Class
Logistic Regression	0.73	0.73	0.73	0.73	0
	0.74	0.73	0.73	0.74	1
	0.73	0.72	0.73	0.73	2
Decision Tree	0.87	0.87	0.87	0.86	0
	0.88	0.87	0.87	0.87	1
	0.87	0.87	0.88	0.87	2
Random Forest	0.93	0.93	0.93	0.93	0
	0.93	0.92	0.93	0.93	1
	0.93	0.93	0.93	0.92	2
Support Vector Machine	0.87	0.87	0.87	0.87	0
	0.86	0.87	0.87	0.86	1
	0.87	0.87	0.86	0.86	2
Naive Bayes	0.59	0.50	0.59	0.48	0
	0.60	0.58	0.58	0.52	1
	0.59	0.58	0.58	0.48	2
k-Nearest Neighbors	0.79	0.80	0.79	0.78	0
	0.80	0.80	0.80	0.79	1
	0.80	0.80	0.79	0.78	2
Gradient Boosting	0.92	0.92	0.92	0.92	0
	0.92	0.92	0.91	0.92	1
	0.92	0.92	0.92	0.92	2
eXtreme Gradient Boosting	0.93	0.93	0.93	0.93	0
	0.94	0.93	0.93	0.93	1
	0.93	0.93	0.94	0.94	2
ACO-XgBoost	0.98	0.97	0.97	0.98	0
	0.97	0.98	0.98	0.98	1
	0.98	0.98	0.97	0.98	2

Table 1. Evaluation of machine learning models' performance across accuracy, precision, recall, and F1-score for different classes

Fig. 2 represents a comparative analysis of the results of various machine learning models using metrics such as accuracy, precision, recall, and F1 score. The proposed model, developed to tackle challenges in soil fertility prediction, shows an excellent improvement in all these areas. The proposed model achieves an outstanding accuracy of 0.98 and surpasses standard models such as Treebag [12], Random Forest (RF) [12, 13], C5.0 [12], GBM [12], KNN [12], CART [12], and SVM [12]. Additionally, it successfully beats models like Ensemble [14], DNN [14], DBN [14], SVM [14], and KNN [14] in terms of precision: 0.98. The model also achieves a recall of 0.98 and an F1 score of 0.98, showing superior performance in correctly identifying true positives compared to the Ensemble, DNN, DBN, SVM, and KNN models.

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Fig. 2. Performance comparison of various machine learning algorithms across multiple evaluation metrics: Accuracy, Precision, Recall, and F1 Score.



4.1 Explanation of Correct and Incorrect Classifications Using Explainable AI (XAI)

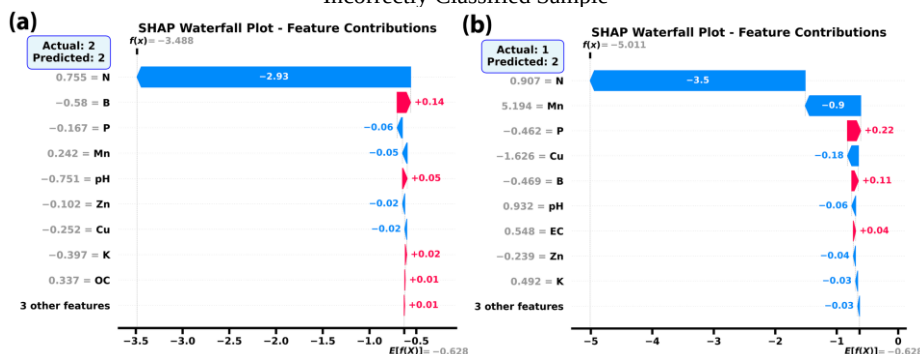
Correct Classification:

The model's prediction aligns with the proper label in Fig. 3 (a), where Actual: 2 and Predicted: 2. The SHAP values for each feature contribute to the final prediction, which is correctly classified as Class 2. The total model output represented as $f(x) = -3.488$, is reached through the sum of individual feature contributions. For instance, N (Nitrogen) contributes with a value of $+0.755$, indicating a strong influence in the correct classification toward Class 2. Other features, such as Mn (Manganese) with a contribution of $+0.242$ and pH with -0.751 , also play roles, although their contributions are smaller. The color gradient indicates each feature's SHAP value, with positive contributions in blue (pushing the output toward Class 2) and negative contributions in red (pulling it away). The cumulative sum of these contributions brings the final prediction within the correct class range, confirming the model's accurate prediction.

Incorrect Classification:

In Fig. 3 (b), Actual: 1, Predicted: 2, the model incorrectly predicts Class 2 instead of Class 1. The total model output, $f(x) = -5.011$, is influenced by individual features. Mn (Manganese), with a strong contribution of $+5.194$, and N (Nitrogen), with $+0.907$, push the prediction toward Class 2, even though the actual label is Class 1. Negative contributions from Cu (Copper) (-1.626) and B (Boron) (-0.469) work to move the prediction away from Class 2 but aren't enough to overcome the positive contributions. The SHAP values show that despite the negative contributions, the combination of features still leads to an incorrect prediction, with the final output closer to Class 2 rather than Class 1.

Fig. 3. SHAP Waterfall Plots for Feature Contributions: (a) Correctly Classified Sample, (b) Incorrectly Classified Sample



5 Conclusion

This work developed a robust soil classification model by using micro and macro soil nutrients to classify soil fertility accurately. The proposed hybrid machine learning approach, integrating ACO with XgBoost, demonstrated superior performance over traditional algorithms, achieving an outstanding accuracy of 98 %. The hyperparameter optimization of XgBoost with ACO significantly enhanced the model's performance. Moreover, XAI techniques provide a valuable understanding of the key soil properties driving the classification process, which ensures greater interpretability and transparency in the decision-making process. This work advances soil fertility classification and contributes to developing more efficient and interpretable machine-learning models in agricultural science, offering practical implications for precision farming and soil management.

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